

PROJECT NAME:

Job Market

PREPARED FOR:

NTI Summer Training

Proposal

Job Market Analytics

01 Overview

The increasing volume of online job postings requires scalable technologies capable of processing large datasets efficiently. This project develops a complete Big Data pipeline using Apache Kafka, Hadoop HDFS, Apache Spark, Apache Hive, Docker, and Power BI. The system ingests, processes, analyzes, and visualizes job market data to provide valuable insights into hiring trends, salary distribution, required skills, and workforce demands.



02 Objectives

- Build a complete Big Data pipeline.
- Ingest streaming data using Apache Kafka.
- Store raw data in Hadoop HDFS.
- Clean and transform data using Apache Spark.
- Analyse the processed data using Apache Hive.
- Visualise insights using Power BI.
- Deploy the entire environment using Docker Compose.



DATA INGESTION

Stream job data using Apache Kafka into Hadoop HDFS.



DATA PROCESSING

Clean, transform, and prepare the dataset using Apache Spark.



DATA ANALYTICS

Analyze the processed data with Hive and visualize insights in Power BI.

Job Descriptions Dataset

The Job Descriptions Dataset contains approximately 1.6 million job postings with 23 attributes, including job titles, companies, locations, salaries, required skills, qualifications, and employment details. It provides a comprehensive source for analyzing hiring trends, workforce demands, and salary distributions using Big Data technologies.

Big Data Pipeline

01

Data Ingestion

Import the CSV dataset into Apache Kafka and stream the data to Hadoop HDFS for scalable storage.

PHASE 1:

Data Collection & Ingestion

02

Data Processing

Use Apache Spark to clean data, remove missing values, convert data types, and extract useful features.

PHASE 2:

Data Cleaning & Processing

03

Data Analysis

Load the cleaned data into Apache Hive and execute SQL queries to analyze hiring trends, salaries, and required skills.

PHASE 3:

Data Analysis

04

Visualization

Connect Hive results to Power BI to build interactive dashboards that present key business insights.

PHASE 4:

Dashboard

Project Summary

This project demonstrates a complete Big Data analytics pipeline using Apache Kafka, Hadoop, Spark, Hive, Docker, and Power BI. The proposed solution efficiently processes large-scale job market data and provides valuable insights into employment trends, salaries, and workforce demands.

DELIVERABLES	TECHNOLOGIES USED	EXPECTED OUTCOMES
• Apache Kafka Pipeline • Hadoop HDFS Storage • Spark ETL Pipeline • Hive Database • Power BI Dashboard • Docker Compose	• Apache Hadoop HDFS • Apache Kafka • Apache Spark • Apache Hive • Docker, Docker Compose • Power BI	• Most Demanded Jobs • Salary Distribution • Hiring Trends • Top Hiring Countries • Most Required Skills • Qualification Analysis

TEAM MEMBERS

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